23.4 A 6.4Tb/s 4.2pJ/b Co-Packaged Optics ASIC with Direct-Drive Integrated TIA and Retimed Segmented Mach-Zehnder Modulator Driver in 7nm FinFET

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Abstract

A 6.4Tb/s co-packaged optics ASIC integrates 64 ingress path direct-drive TIAs and 64 egress path retimed segmented Mach-Zehnder modulator drivers in 7nm FinFET. The 7nm ASIC is 3D packaged with a photonics IC (PIC) and achieves an energy efficiency of 4.2pJ/b. At 106.25Gb/s, its PAM-4 TIA has a sensitivity better than -11dBm and a BER floor better than 1E-9. The transmitted PAM-4 optical eye achieves a TDEQ of 1.48dB and an extinction ratio of 4.57dB.

Optical interconnects are gaining importance for the high-bandwidth communication needs of the scale-out AI clusters, replacing the pluggable modules. Co-packaged optics (CPO) is a heterogeneous integration technology to address the bandwidth and power consumption challenges by enabling scaling through integration. The shortened chip-to-chip interconnects between a photonic integrated IC (PIC) and a networking ASIC, reduces complexity, power consumption and signal integrity issues of otherwise lengthy interconnects between optics and electronics. This work presents a 6.4Tb/s retimer ASIC for CPO applications. Figure 23.4.1 shows the monolithic integration of 64 egress and ingress lanes of 106.25Gb/s PAM-4 channels. These lanes directly interface the line side on a PIC driving 64 segmented Mach-Zehnder modulators (MZM) [1] for TX and interface 64 photodiodes (PD) for RX [2]. An AFE followed by a CDR retimes and deserializes the host egress path incoming data. The CDR runs off the 4t clock from the TX PLL. The deserialized data bus feeds the MZM driver through the TX DSP, where the thermometer coding of the PAM-4 signal allows the concurrent drive of two segments with the MSB data. The ingress path implements a DSP-adapted TIA. The TIA output directly drives the host ASIC through an organic substrate. Furthermore, 320 monitoring photodiode readouts and 320 heater DACs are integrated within the ASIC for the optical sense and control of the PIC.

The host interface in the egress path is shown in Fig. 23.4.2. The input signal is capacitively coupled to the inputs of a continuous-time linear equalizer (CTLE) through a calibrated termination, which utilizes bandwidth extension t-coils for the input ESD. The CTLE, whose input common-mode is set by a programmable DAC, drives a variable-gain amplifier (VGA), which then drives the 4× time-interleaved sampler. The CTLE and VGA both utilize resistor calibration and t-coils for bandwidth extension.

The 4 sampling phases of the CDR are generated off the TX PLL clk4t clocks using a clkgen block. Within the clkgen block, a digitally adapted DLL-based quadrature clock generator (IQgen) drives a phase interpolator (PI) followed by a second IQgen producing 0, 90, 180 and 270 degrees sampling phases. The 4 slicer banks feed data and error information to a de-serializer, which in turn produces 128 data and 64 error bits at clk64t update rate to the TX DSP. The TX DSP contains CDR adaptation where the clock recovery loop is implemented. It eventually feeds the PAM-4 MSB and LSB data to the line side. A slicer level adaptation algorithm within the TX DSP adjusts the 4 threshold levels of the slicers within each sampler bank using sixteen 9b DACs.

The line side interface of the egress path is shown in Fig. 23.4.3. The serializer first translates the 64b retimed parallel MSB and LSB data from the TX DSP into 8b data streams. These data buses in the 8t clock domain are then passed to 3 driver segments, with the first segment assigned to the LSB and the last two to the MSB. The TXPLL 4t clock is distributed along the 3 MZM driver segments. Each segment's local clock generator corrects for clock distortion and generates 4 phases by means of a quadrature clock generation block. The 8t and 4t calibrated clocks from the IQ gen drive the last two serializer stages of the segment driver. The 4:1 mux output is translated into a 1.5V NRZ pattern by means of a two-stage level shifter pre-driver that drives the PMOS and NMOS gates of the stacked output stage. Pre-driver and cascode voltage levels are generated within the segment by means of level generation LDOs and DACs, respectively.

The ingress path consists of a direct drive TIA as shown in Fig. 23.4.4. On the line side, the PD anode interfaces a current-to-voltage converter (i2v). An input coil L_{in} extends the bandwidth by resonating the input parasitic capacitance. The i2v core is built around an inverter with a feedback resistor R_{fb} and regulated bias current. The i2v stage drives a first VGA1 pseudo-differentially after which the signal path through VGA2, 3 and the output driver remain fully-differential. The VGAs fan out towards the driver and incorporate bandwidth extension by means of t-coils and make use of 8b adjustable calibrated load resistors. The gain adjustment is through a digital gain adjustment code. The output driver uses 5b termination calibration and t-coils for bandwidth extension and driving the ESD protection and output bumps. The TIA differential output drives the host switching ASIC directly through an organic substrate. Each lane's TIA is digitally adapted by means of a local DSP, which drives the VGA gain control word for amplitude regulation, as well as a DC cancellation DAC connected to the TIA input. Adaptation information is provided to the DSP by means of an adaptation frontend. Each VGA is capable of ~6dB gain with gain programmability steps of 0.2dB. The TIA has a nominal input current noise density of 15pA/√Hz while consuming less than 50mW of power. Furthermore, the ingress path integrates local PTAT bias and supply regulation within each TIA channel.

The CPO ASIC is fabricated in a 7nm FinFET technology (Fig. 23.4.7). The monolithic die integrates 64 egress and ingress lanes, TXPLLs, auxiliary common circuits, ADC and DAC components. The 6.4Tb/s optical engines are built by 3D packaging of the 7nm ASIC and the PIC silicon. Multiple engines are co-packaged with a host networking ASIC for scalable throughputs. The performance of the direct drive TIA ingress path is shown in Fig. 23.4.5, where the TIA output has been equalized by the host networking ASIC. The BER vs input OMA curves across 512-channels of a 51.2Tb/s CPO system (8×6.4Tb/s optical engines) is shown across the 4 utilized wavelengths. The TIA achieves an average sensitivity better than -11dBm (IEEE802.3 BER of 2.4E-4 for 106.25Gb/s PAM-4) and an average BER floor better than 1E-9. At 106.25Gb/s PAM-4 data rate, the egress path has a pre-FEC BER of better than 1E-9 where more than 99.98% of channels are error free. An eye-scan of the egress path's CDR, captured by the TX DSP, shows an open PAM-4 eye in Fig. 23.4.6. The optical PAM-4 eye (Fig. 23.4.6) for a PRBS-13 pattern is measured by a DCA applying an IEEE 802.3bs reference RX equalizer with a 5-tap FFE. The TX achieves a transmitter and dispersion eye closure quaternary (TDEQ) of 1.48dB and an outer extinction-ratio (ER) of 4.57dB and a ratio-level mismatch (RLM) of 0.958. Comparison to state-of-the-art [3 – 6] optical interface transceivers (Fig. 23.4.6) shows the 6.4Tb/s CPO ASIC achieves the highest level of integration and throughput with ×64 106.25Gb/s PAM-4 lanes at a competitive energy efficiency of 4.2pJ/b.

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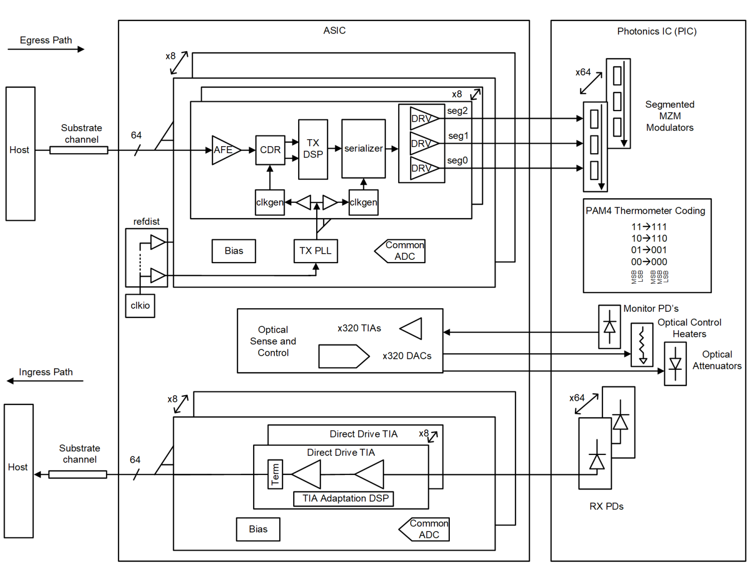


Figure 23.4.1: 64Tb/s ASIC and ingress and egress paths to the host and photonics IC (PIC).

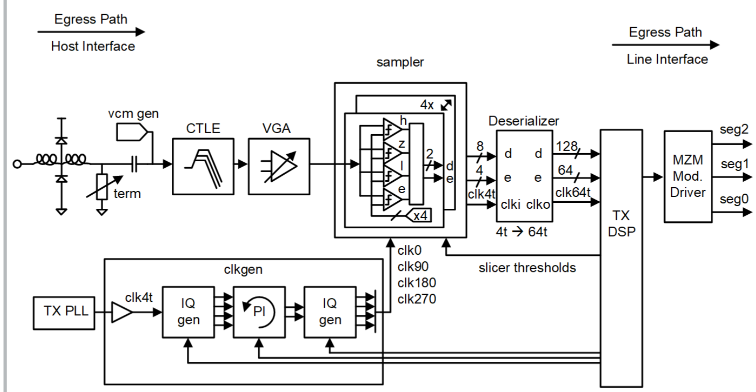


Figure 23.4.2: Egress path host interface AFE, CDR sampler and deserializer.

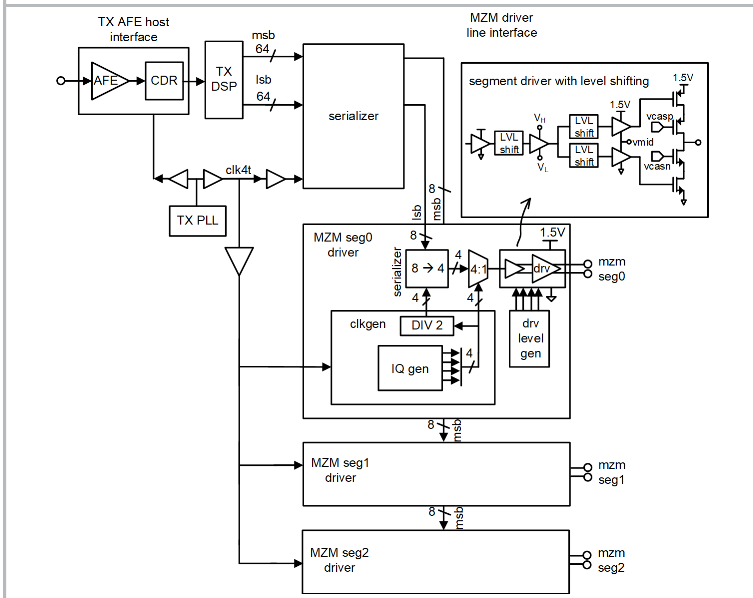


Figure 23.4.3: Egress path line interface, serializer, clock gen and MZM modulator driver.

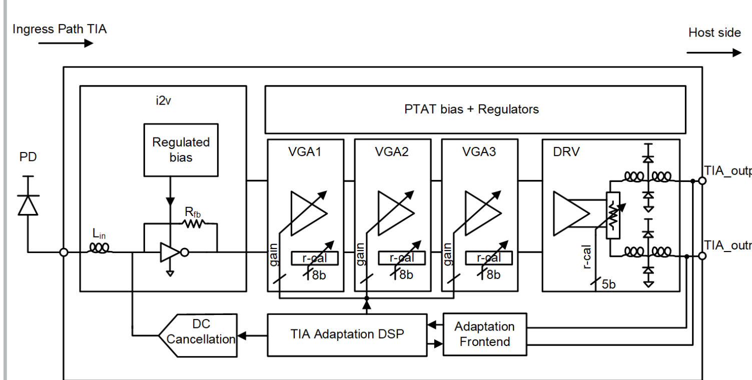


Figure 23.4.4: Ingress path direct drive TIA.

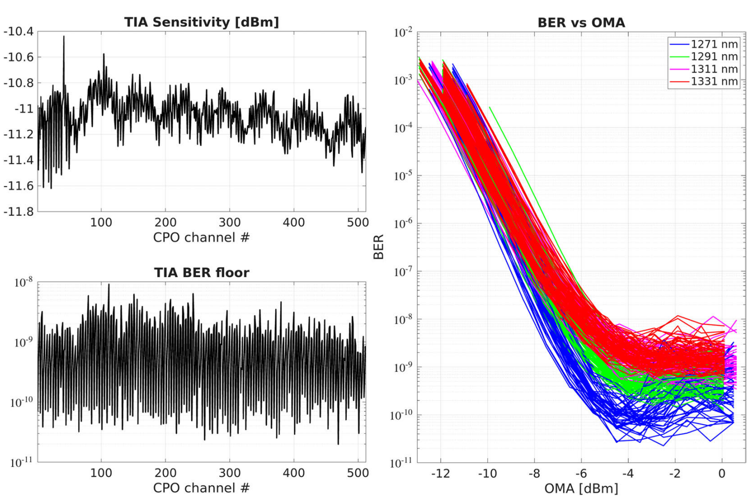
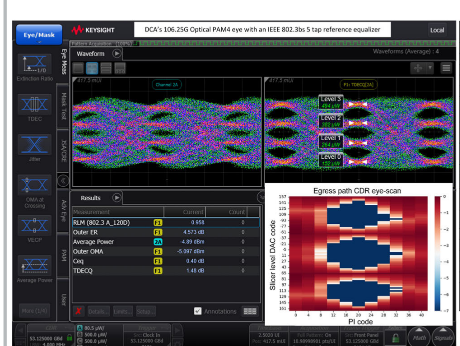


Figure 23.4.5: Direct drive TIA 106.25Gb/s PAM-4 sensitivity, BER floor and BER vs input OMA across 512 channels of a 51.2Tb/s CPO system comprising eight 6.4Tb/s optical engines.



Parameter	This work	[3] ISSCC25	[4] ISSCC25	[5] ISSCC24	[6] ISSCC24	[1] ISSCC22	[2] ISSCC23
Technology	7nm	5nm	22nm	5nm	22nm	28nm	16nm
Area [mm2]	105	1.63	0.19+0.19	0.64+0.72	0.19+0.13	0.62	0.068
Engine type	CPO	MCM*	MCM*	MCM*	MCM*	CPO	MCM*
# of lanes	64	4	1	4	4	1	1
Transceiver	TX + RX	TX + RX	TX + RX	TX + RX	TX	TX	RX
Data rate Modulation	106Gbp PAM4	212Gbps PAM4	128Gbps PAM4	212Gbps PAM4	64Gbps NRZ	100Gbps PAM4	106Gbps PAM4
ASIC throughput	6.4Tbps	0.8Tbps	0.1Tbps	0.8Tbps	0.2Tbps	0.1Tbps	0.1Tbps
Energy eff. [pJ/b]	4.2	2.76 + 4.4	0.31 + 0.9	2.69	1.3+0.66	2.4	1.016
TX / Modulator	seg. MZM	MZM	VCSEL		VCSEL	seg. MZM	
Integrated TIA + DRV	Yes	No	No	No	No	No	No
TIA input ref. Noise pA/rt(Hz)	15						7
TX BER	1.00E-09	1.80E-10	1.00E-06		1.00E-12		
TX RLM	0.958	0.994		0.98			
TX ER	4.57 dB						
TX TDEQ	1.48 dB	2.04 dB				1.53 dB	
RX BER floor	1.00E-09						< 1E-13
RX Sensitivity	-11dBm		-6.2dBm 108G				-14dBm

*MCM: Multi-chip module

Figure 23.4.6: TX PAM-4 optical eye at 106.25Gb/s, egress CDR eye-scan and comparison with state-of-the-art.

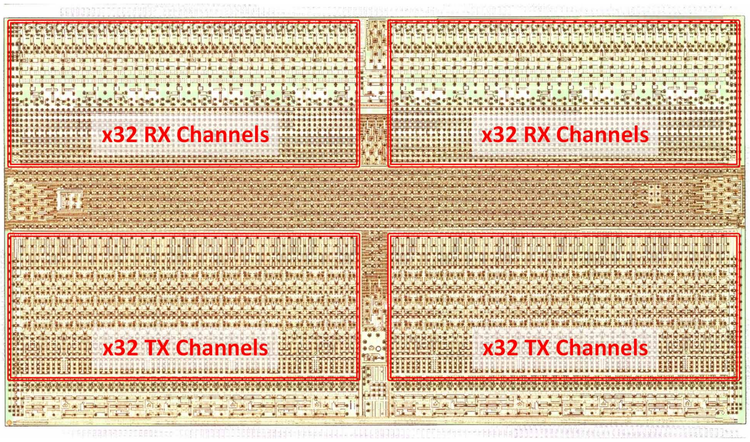


Figure 23.4.7: 6.4Tb/s co-packaged optics ASIC in 7nm FinFET.

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